

THE FREEDOM ENGINE 4 PILLARS ARCHITECTURE

Empirical Comparison: Live Engine (2022–2026) vs. Multi-Cycle Historical Proxy (2013–2026)

1. Executive Summary & Foundational Mandate

The **Freedom Engine 4 Pillars Architecture** is an institutional liability-driven investment framework engineered specifically for leveraged, positive-carry wealth accumulation. Traditional leveraged accounts routinely succumb during market drawdowns to three structural failure modes:

- **Interest Rate Duration Risk:** Fixed-coupon bonds (e.g., BND, AGG) drop sharply in capital value when benchmark rates rise, precisely when margin borrowing costs increase.
- **Synthetic Yield Decay:** Aggressive single-stock synthetic option funds (e.g., YieldMax) distribute their own capital, triggering destructive multi-year NAV erosion.
- **The Drawdown Ratchet:** Market drawdowns expand debt Loan-to-Value (LTV) non-linearly, forcing untimely liquidation at market troughs.

By deploying four segregated economic engines across floating collateral, productive index call spreads, quality equity compounding, and multi-asset trend following, the architecture generates equity-like returns (**13.03%–14.24% CAGR**) with **46%–61% lower volatility** than the S&P 500, a robust **1.40× positive asymmetry ratio**, and a self-funding **7.00% organic cash yield** that eliminates forced asset liquidations.

2. The 4-Pillar Structural Architecture

PILLAR & ALLOCATION	LIVE FUND (PROXY)	ASSET CLASS & EXPOSURE	CORE ECONOMIC FUNCTION & RISK BUFFER
Pillar 1: Collateral Floor 35.0% Total	JAAA: 20.0% (FLOT) SRLN: 15.0% (BKLN)	AAA CLOs & Senior Secured Floating Bank Loans	Zero Interest Rate Duration: Coupons adjust upward automatically with SOFR/Prime. Anchors portfolio collateral with minimal drawdowns (-2.6% to -4.0%).
Pillar 2: Productive	SPYI: 20.0%	Active OTM Index Call Spreads (S&P 500 &	Autonomous Debt Servicing: Generates 11%+ distribution yields via out-of-the-money options

PILLAR & ALLOCATION	LIVE FUND (PROXY)	ASSET CLASS & EXPOSURE	CORE ECONOMIC FUNCTION & RISK BUFFER
Carry 35.0% Total	(XYLD) JEPQ: 15.0% (QYLD)	Nasdaq-100)	and credit call spreads, permitting NAV growth (+4.3% to +8.4%).
Pillar 3: Compounder 20.0% Total	PVAL: 20.0% (VTV)	Concentrated Large-Cap Value Equities	Collateral Base Expansion: Compounds net equity value at high risk-adjusted rates (Sharpe 2.59, 23.5% 3Y return) with shallow drawdowns (-16.6%) to dilute borrowing LTV.
Pillar 4: Crisis Alpha 10.0% Total	DBMF: 10.0% (AQMIX)	Systematic Trend-Following Managed Futures (SG CTA)	Non-Correlated Market Hedge: Trades long/short across currencies, commodities, rates, and equity futures. Displays near-zero equity correlation (0.14 beta).

3. Empirical Head-to-Head Comparison: Live Engine vs. 13-Year Proxy

Extracted from rigorous *PortfoliosLab* analytics, this comparison contrasts the modern live vehicle mix against its 13-year institutional proxy and the broad S&P 500 Index benchmark.

DIMENSION & METRIC	FE 4 PILLARS (LIVE ENGINE)	FE 4 PILLARS PROXY (13-YEAR)	S&P 500 BENCHMARK	INSTITUTIONAL PERFORMANCE ANALYSIS
Core Holdings	JAAA, SRLN, SPYI, JEPQ, PVAL, DBMF	FLOT, BKLN, XYLD, QYLD, VTV, AQMIX	S&P 500 Total Return (SPX / SPY)	Live uses modern OTM call spreads; Proxy uses classic ATM buy-writes.
Analyzed Window	2022-08-30 to 2026-09-11 (~4.0 Years)	2013-12-12 to 2026-09-11 (~12.75 Years)	Full Period Match	Proxy covers 2015 energy crash, 2018 hikes, 2020 COVID, & 2022 inflation.
1-Year Return	15.95%	15.66%	16.29%	Both models captured ~97%–98% of the equity bull run with half the risk.
3-Year Ann. Return	14.24%	12.00%	19.72%	Live outpaces proxy by +2.24%/yr due to SPYI/JEPQ out-of-the-money call design.
All-Time Ann. Return	13.03%	7.11%	8.11%	Live outgained S&P by +4.92%/yr ; Proxy compounded steadily through ZIRP.
Annualized Volatility (1Y)	6.96%	4.99%	12.85%	46%–61% volatility suppression protecting collateral from forced margin calls.
Market Beta / Alpha	0.49 / +4.26%	0.45 / +1.40%	1.00 / 0.00%	Extracts full equity market returns on less than half the systematic beta.
Upside / Downside Capture	51.46% / 36.72%	42.29% / 41.71%	100.0% / 100.0%	Live achieves a 1.40× asymmetry ratio , cutting downside participation to ~37%.

DIMENSION & METRIC	FE 4 PILLARS (LIVE ENGINE)	FE 4 PILLARS PROXY (13-YEAR)	S&P 500 BENCHMARK	INSTITUTIONAL PERFORMANCE ANALYSIS
Max Drawdown (All-Time)	-10.85% (2025)	-22.80% (2020)	-56.78% (All-time) -33.92% (10Y)	Proxy proves survival through COVID-19; Live held pullbacks strictly to -10.85%.
Sharpe Ratio (1Y / All-Time)	2.30 / 1.62	3.15 / 0.86	1.27 / 0.47	Top-tier risk efficiency (90th–97th percentile rank in PortfoliosLab peer group).
Sortino Ratio (1Y / All-Time)	3.22 / 2.30	4.57 / 1.17	1.81 / 0.76	Substantial downside-risk efficiency, minimizing destructive drawdown pain.
TTM Distribution Yield	7.00%	6.00%	~1.30%	Generates 5× the organic cash flow of broad equities to service borrowing debt.
Expense Ratio / M1 Status	0.53% / 100% on M1	0.47% / 100% on M1	—	Both models execute cleanly on M1 Finance without illiquid vehicle restrictions.

4. Deconstructing the Return Spread: Why the Live Engine Outperforms the Proxy

The Live Engine compounded at **13.03%–14.24%** while the 13-Year Proxy annualized at **7.11%–7.87%**. This performance spread illustrates two major advancements in vehicle design:

1. OUT-OF-THE-MONEY (OTM) CALL SPREADS VS. AT-THE-MONEY (ATM) TRUNCATION

Legacy covered call funds like XYLD and QYLD sell 100% at-the-money options. During market recoveries, ATM options cap upside immediately at the strike, forcing the fund to absorb subsequent drawdowns with limited rebound. Modern funds like SPYI (OTM call spreads) and JEPQ (dynamic OTM equity-linked notes) preserve upside participation, achieving an **upside capture of 51.46%** vs. the proxy's 42.29%.

2. MONETARY POLICY & FLOATING COLLATERAL NORMALIZATION

Between 2014 and 2021, the Federal Reserve pinned benchmark rates near 0% (ZIRP). During that era, floating credit instruments (FLOT, BKLN) yielded only 1.5%–3.0%. In the current normalized rate regime, benchmark SOFR provides an organic 6.5%–8.0% yield on floating credit (JAAA, SRLN), substantially enhancing baseline positive carry without adding duration risk.

5. Structural Mechanics: Why the Architecture Works Under Leverage

The FE 4 Pillars operates as an engineered balance sheet designed to survive and compound under floating margin liabilities:

Mechanism 1: The 1.40× Asymmetry Ratio Halts the "Drawdown Ratchet"

In any leveraged portfolio, drawdowns expand borrowing Loan-to-Value (LTV) non-linearly according to:

LEVERAGE EXPANSION FORMULA

$$extStressedLTV = \frac{extInitialLTV}{1 - extDrawdown}$$

When a standard 1.0-beta portfolio experiences a 30% drawdown, an initial 25% borrowing LTV expands to:

$$extStressedLTV = \frac{0.25}{1 - 0.30} = \frac{0.25}{0.70} = \mathbf{35.71\%}$$

This crosses common broker maintenance limits, triggering automated margin calls and forced liquidations at the market bottom. Because the Live Engine captures **51.46% of market upside against only 36.72% of downside**:

ASYMMETRY RATIO CALCULATION

$$extAsymmetryRatio = \frac{extUpsideCapture}{extDownsideCapture} = \frac{51.46\%}{36.72\%} = \mathbf{1.401imes}$$

During the deepest market pullback since 2022, the live portfolio suffered only a **-10.85% drawdown**. Stressed leverage drifted from 25.00% to just **28.04%**—leaving a massive safety cushion below the 35% maintenance threshold.

Mechanism 2: Floating Asset-Liability Matching (Zero Duration Risk)

Margin borrowing creates a *floating-rate liability* tied to benchmark rates (SOFR or Prime + broker margin spread). Allocating **35% to pure floating credit** (JAAA and SRLN) matches the interest rate sensitivity of the debt.

Because loan coupons reset every 30 to 90 days, cash flow increases automatically when financing costs rise, neutralizing interest rate duration risk.

Mechanism 3: Stressed Cash-Interest Coverage Exceeds the 3.0× Institutional Mandate

Generating an organic **7.00% portfolio distribution yield**, incoming cash flow easily covers debt service across both baseline and stressed borrowing conditions:

CASH INTEREST COVERAGE FORMULA

$$extCoverage = \frac{extPortfolioDistributionYield}{extLTVimesextBorrowingRate}$$

BORROWING SCENARIO	BORROWING RATE	DEBT SERVICE HURDLE (AT 25% LTV)	PORTFOLIO YIELD	CASH INTEREST COVERAGE	SOLVE VERDICT
Base Floating Rate	6.00%	$0.25imes6.00\% = 1.50\%$	7.00%	$\frac{7.00\%}{1.50\%} = 4.67imes$	PASS (Abundant Coverage)
Stressed Rate Spike	8.00%	$0.25imes8.00\% = 2.00\%$	7.00%	$\frac{7.00\%}{2.00\%} = 3.50imes$	PASS (Clear \$1ge 3imes\$ Mandate)

Mechanism 4: True Factor Decoupling via Trend-Following Crisis Alpha

The 10% allocation to **DBMF** provides genuine non-correlation by taking long/short positions across currencies, commodities, rates, and global equities. Empirical correlation data proves this independence: DBMF exhibits a **0.14 correlation to the S&P 500**, **0.03 to JAAA**, and **0.05 to SRLN**, stepping in as a counter-cyclical stabilizer during macro sell-offs.

6. Multi-Cycle Stress Testing: Historical Crises in the 13-Year Proxy (2013–2026)

The 13-year proxy backtest covers four severe market shocks, demonstrating how the architecture behaves across diverse economic regimes:

HISTORICAL CRISIS EVENT & DATES	S&P 500 MAX DD	PROXY MAX DD	STRESSED LTV (25% START)	STRESSED LTV (30% START)	SOLVENCY VERDICT & MECHANICAL BEHAVIOR
COVID-19 Liquidity Shock (2020-02-21 to 2020-03-23)	-33.92%	-22.80%	32.38%	38.86%	SURVIVED AT 25% LTV (2.62% buffer below 35% margin call). FAILED AT 30% LTV (Forced liquidation at bottom).
2018 Fed Rate Hikes / Q4 Rout (2018-10-04 to 2018-12-24)	-19.78%	-11.38%	28.21%	33.85%	Comfortable solvency buffer (6.79% cushion). Trend following and floating debt protected capital.
2022 Inflation & Rate Shock (2022-04-21 to 2022-09-30)	-25.43%	-9.88%	27.74%	33.29%	Floating credit floor completely shielded collateral while traditional 60/40 portfolios collapsed -16%+.
2015–16 Energy & Credit Rout (2015-05-22 to 2015-08-25)	-14.16%	-6.40%	26.71%	32.05%	Negligible collateral drift; cash flow continued self-servicing borrowing debt smoothly.

7. Mathematical Proof of the 25% LTV Borrowing Ceiling

During the March 2020 COVID crash, the proxy portfolio sustained an all-time maximum drawdown of **-22.80%**. Evaluating borrowing leverage through this liquidity shock establishes the strict mathematical boundary for portfolio safety:

$$extStressedLTV = \frac{extInitialLTV}{1 - 0.2280} = \frac{extInitialLTV}{0.7720}$$

- **Case A: Initial 25.0% LTV**

$$extStressedLTV = \frac{0.25}{0.7720} = \mathbf{32.38\%}$$

Outcome: The portfolio absorbed the fastest 30%+ drop in modern financial history with a **2.62% safety buffer** beneath standard 35% broker maintenance thresholds. No assets were liquidated, and the portfolio achieved new all-time highs by December 2020.

- **Case B: Initial 30.0% LTV**

$$extStressedLTV = \frac{0.30}{0.7720} = \mathbf{38.86\%}$$

Outcome: Leverage breached the 35% maintenance threshold, triggering automated broker margin calls and forced liquidations at the market trough.

Non-Negotiable Solvency Rule: Initial borrowing leverage must never exceed **25.0% LTV**. This guarantees that even a historic liquidity crisis leaves a safe operating buffer.

8. Dynamic Coverage Engineering: Adjusting Pillars for Higher Debt Service

When macroeconomic conditions tighten—such as margin borrowing rates surging to 9%–10% or target leverage expanding toward 30% LTV—the portfolio's **Debt Service Coverage Ratio (DSCR)** must be dynamically widened. The mathematical lever is the portfolio's organic yield:

DEBT SERVICE COVERAGE RATIO FORMULA

$$extDSCR = \frac{extPortfolioDistributionYield}{extLTVimesextBorrowingRate}$$

Because **Pillar 3 (PVAL)** yields only **0.89%**, it contributes just 2.6% of the portfolio's cash flow despite occupying 20% of capital. PVAL serves as the primary *"liquidity donor."* By trimming PVAL to fund higher-yielding productive carry (**Pillar 2**) or shifting internal weights within the floating collateral floor (**Pillar 1**), cash flow can be increased without sacrificing downside collateral protection.

PORTFOLIO SCENARIO	PILLAR ALLOCATIONS (P1 / P2 / P3 / P4)	WEIGHTED YIELD	3Y ANN. RETURN	COVERAGE @ 25% LTV (8% STRESS)	COVERAGE @ 25% LTV (10% SPIKE)	COVERAGE @ 30% LTV (8% STRESS)
Baseline (Growth Focus)	P1: 35% (JAAA 20, SRLN 15) P2: 35% (SPYI 20, JEPQ 15) P3: 20% (PVAL) P4: 10% (DBMF)	7.00%	14.24%	3.50×	2.80×	2.92×
Scenario 1: 30% LTV Cushion	P1: 35% (JAAA 20, SRLN 15) P2: 45% (SPYI 25, JEPQ 20) P3: 10% (PVAL) P4: 10% (DBMF)	7.87%	13.59%	3.93×	3.15×	3.28×
Scenario 2: Rate Spike (9%–10%)	P1: 35% (JAAA 15, SRLN 20)	7.99%	13.64%	3.99×	3.20×	3.33×

Portfolio Scenario	Pillar Allocations (P1 / P2 / P3 / P4)	Weighted Yield	3Y Ann. Return	Coverage @ 25% LTV (8% Stress)	Coverage @ 25% LTV (10% Spike)	Coverage @ 30% LTV (8% Stress)
	P2: 45% (SPYI 25, JEPQ 20) P3: 10% (PVAL) P4: 10% (DBMF)					
Scenario 3: Max Deleveraging	P1: 35% (JAAA 15, SRLN 20) P2: 50% (SPYI 25, JEPQ 25) P3: 5% (PVAL) P4: 10% (DBMF)	8.51%	13.46%	4.25×	3.40×	3.55×

Strategic Scenario Dynamics & Operational Triggers

- **Scenario 1 (Operating at 28%–30% Target LTV):** If higher leverage is drawn, the 8.0% interest hurdle rises from 2.00% to 2.40%. Reallocating 10% from PVAL into SPYI (+5%) and JEPQ (+5%) boosts yield to **7.87%**, restoring coverage to **3.28×**.
- **Scenario 2 (Borrowing Rate Shock at 9.0%–10.0%):** When benchmark rates spike, tilt Pillar 1 internally toward SRLN (7.27% yield) over JAAA (4.89% yield). This captures wider floating credit spreads, pushing yield to **7.99%** and holding coverage above **3.20×** even under a 10.0% debt rate.
- **Scenario 3 (Rapid Margin Deleveraging Mode):** Expanding Pillar 2 to 50% pushes yield to **8.51%**. Against a 6.0% borrowing cost (1.50% hurdle at 25% LTV), the portfolio produces **5.67× coverage** and leaves **+7.01% in net cash flow** to pay down margin principal automatically via M1 Smart Transfers.

NON-NEGOTIABLE ARCHITECTURAL GUARDRAILS

1. **Never Reduce Pillar 4 (DBMF) Below 10.0%:** Sacrificing DBMF for yield eliminates crisis alpha (0.14 beta to S&P). Without it, drawdowns deepen during stock/bond routs.
2. **Maintain Pillar 1 at ≥ 35.0%:** Floating credit eliminates rate duration. Cutting Pillar 1 increases mark-to-market volatility and risks margin calls.
3. **Pillar 3 (PVAL) is the Floating Swing Factor:** Dial PVAL down to 5%–10% during high-rate regimes to maximize debt coverage; restore to 20% when borrowing rates drop.

9. Implementation Blueprint: M1 Finance Sub-Pie Configuration

To deploy the Freedom Engine on M1 Finance, structure the account into four segregated sub-pies:

M1 SUB-PIE SLICE	SUB-PIE WEIGHT	UNDERLYING TICKER	SLICE TARGET	MASTER TARGET	OPERATIONAL ROLE & EXPOSURE
1. Collateral Floor	35.0%	JAAA	57.0%	20.0%	Janus Henderson AAA CLO ETF (SOFR Floater) SPDR Blackstone Senior Loan ETF
		SRLN	43.0%	15.0%	
2. Cash Carry	35.0%	SPYI	57.0%	20.0%	NEOS S&P 500 High Income (OTM Call Spreads) JPMorgan Nasdaq Equity Premium Income
		JEPQ	43.0%	15.0%	
3. Compounder	20.0%	PVAL	100.0%	20.0%	Putnam Focused Large Cap Value ETF (NAV Growth Engine)
4. Crisis Alpha	10.0%	DBMF	100.0%	10.0%	iMGP DBi Managed Futures Strategy ETF (CTA Trend Hedge)
Total Freedom Engine	100.0%	—	—	100.0%	7.00% Organic Yield 14.24% 3Y Return Beta 0.49

10. Execution Protocols: Cash Flow Sweeping & Rebalancing Rules

- The Strict 25.0% Initial LTV Cap:** Never initiate margin borrowing beyond 25.0% of total portfolio value (\$25,000 borrowed on a \$100,000 portfolio). This maintains a minimum 10% market-drop buffer before reaching 28% LTV and safely absorbs a 2020-style -22.8% shock without breaching 35% maintenance thresholds.
- Automated Smart Transfer Distribution Sweeping:** The portfolio generates 7.00% in organic cash flow annually. In M1 Finance, configure automated Smart Transfer rules to route distributions dynamically:
 - Normal Regime (LTV < 25.0%):* Auto-reinvest incoming distributions into underweighted sub-pies, systematically buying whichever pillar is lagging.

- *Stressed Regime (LTV > 28.0%)*: Automatically route all incoming cash distributions directly into paying down margin balance until LTV drops back below 25.0%. This creates an autonomous, self-healing deleveraging mechanism during market corrections without selling depressed assets.

3. **Cash-Flow Rebalancing Bands ($\pm 4\%$ Drift)**: Avoid selling appreciated assets to rebalance, which triggers capital gains taxes. Instead, establish a $\pm 4.0\%$ absolute tolerance band for each sub-pie. Direct monthly dividend distributions and new deposits to top off underweighted slices, achieving tax-free rebalancing through cash flow.